

Q. show what languages are generated by the given CFG (Context Free grammar) in each case.

$$S \rightarrow aSb \mid bSa$$

(4 marks)

$$S \rightarrow SS \mid bS \mid a\lambda$$

⇒ ① $S \rightarrow aSb \mid bSa$

$$S \rightarrow aSb$$

means : a at beginning
b at end

if inner S stops:

$$\text{for } S \rightarrow aSb, S \rightarrow ab$$

$$S \rightarrow bSa, S \rightarrow ba$$

$$\text{if we take } S \rightarrow aSb$$

$$\rightarrow abSab$$

$$\rightarrow abab$$

$$\text{if we take } S \rightarrow bSa$$

$$\rightarrow baSba$$

$$\rightarrow baba$$

observe the pattern :
every derivation adds:
one 'a' and one 'b'
∴ number of a's = number of b's

→ as number of a's = number of b's,

Language Generated :

$$L = \{w \mid n_a(w) = n_b(w)\}$$

here, w means string

$n_a(w) \rightarrow$ no of a's in string w

$n_b(w) \rightarrow$ no of b's in string w

② $S \rightarrow SS \mid bS \mid a\lambda$

since $a\lambda = a$,

Grammar is :

$$S \rightarrow SS \mid bS \mid a$$

(i) $S \rightarrow SS$

(joins two strings)

(ii) $S \rightarrow bS$

(adds b at beginning)

(iii) $S \rightarrow a$ (terminates derivation)

$$\Rightarrow S \rightarrow a$$

$$\therefore \text{string} : a \text{ ————— } \textcircled{1}$$

$$\Rightarrow S \rightarrow bS$$

$$\rightarrow ba \text{ (because } S \rightarrow a)$$

$$\therefore \text{string} : ba \text{ ————— } \textcircled{2}$$

$$\Rightarrow S \rightarrow bS$$

$$\rightarrow bbs \text{ (because } S \rightarrow bS)$$

$$\rightarrow bba$$

$$\therefore \text{string} : bba \text{ ————— } \textcircled{3}$$

$$\Rightarrow S \rightarrow SS$$

$$\rightarrow aS \text{ (because } S \rightarrow a)$$

$$\rightarrow abS \text{ (because } S \rightarrow bS)$$

$$\rightarrow aba \text{ (because } S \rightarrow a)$$

$$\therefore \text{string} : aba \text{ ————— } \textcircled{4}$$

$$\Rightarrow S \rightarrow SS$$

$$\rightarrow bSS$$

$$\rightarrow baS$$

$$\rightarrow baa$$

$$\therefore \text{string} : baad \text{ ————— } \textcircled{5}$$

from $\textcircled{1}, \textcircled{2}, \textcircled{3}, \textcircled{4}$ & $\textcircled{5}$, pattern : every string contains any no of a's or b's in the beginning and ends with 'a'.

$\therefore L = \{ \text{All strings over } \{a, b\} \text{ that end with 'a'} \}$

$$\text{A.E.} = (a+b)^* a$$

Q. For each of the following, describe the language generated by given CFG : (4 Marks)

$$S \rightarrow SaS \mid b$$

$$S \rightarrow aT \mid bT \mid \Lambda$$

$$T \rightarrow aS \mid bS$$